

Design Basis Report - Critical Infrastructure Systems

Project: PRJ-MUM-2026 | Code: DBR-001

Design Philosophy: Facility is designed to Uptime Institute Tier III equivalent standards with concurrently maintainable electrical and mechanical distribution paths.

Redundancy Strategy: N+1 redundancy is mandated at the individual equipment level across UPS, chiller, and CRAH systems; 2N is not required for this phase.

Design Margin: All critical electrical and mechanical systems must be sized with a minimum 20% headroom above calculated peak IT load to accommodate future scaling.

Parameter	Value	Notes / Tolerance
Target Tier Level	Tier III Equivalent	Uptime Institute framework
Redundancy Level	N+1	Equipment level, this phase
Design Margin	20%	Above calculated peak IT load
Design Life	20 years	Core electrical/mechanical infra